Metric-Semantic Pose Estimation of PC Back-Panel Components

CP260 Robotic Perception — Final Project Report

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May 2026

Abstract

We present a pipeline for estimating the 6-DoF poses of physical components on the back panel of a desktop PC using 16 posed RGB images and pre-calibrated camera intrinsics. For each target component, we manually annotate 2D bounding boxes on the two most informative frames and triangulate a dense grid of correspondences across view pairs to recover a 3D point cluster. An Oriented Bounding Box (OBB) is fitted using PCA augmented with back-panel surface normal estimation and physical depth priors. For the generalization task—where new, previously unseen parts must be localized at evaluation time—we integrate Grounding DINO, a zero-shot open-vocabulary object detector that automatically produces the 2D annotations needed by the same triangulation backend. Validated against a provided VGA socket ground truth, the baseline system achieves a center localization error of approximately 0.6 mm.

1. Problem Description

The task is to build a metric-semantic 3D representation of a desktop PC scene from a set of 16 posed RGB images. The dataset consists of frames captured while orbiting a PC tower placed on a filing cabinet in a laboratory environment. Each image is accompanied by a 4×4 camera-to-world (c2w) transformation matrix stored in `poses.json`, and calibrated camera intrinsics are provided separately in `intrinsic.json`.

The primary objective is to estimate the pose of two specific components on the PC back panel:

- **Power socket —** the IEC C14 inlet at the bottom of the power supply unit
- **Ethernet socket —** the RJ-45 port on the motherboard I/O shield

Each pose is represented as an Oriented Bounding Box (OBB) defined by a 3D center vector, a half-extent vector encoding the physical dimensions of the connector, and a 3×3 rotation matrix. A bonus evaluation requires the system to generalize to arbitrary new parts specified only at test time. The dataset also includes a VGA socket ground truth in `sample_answers.json`, which we used as a validation reference during development.

Working with only 16 images introduces several practical difficulties. The back panel occupies a small portion of the wide-angle frame in most shots, making individual connectors appear at a scale of only a few dozen pixels. The panel surface is dark and nearly textureless, ruling out standard feature-matching based dense reconstruction. The sparse image set also means that stereo baselines between views vary significantly, affecting triangulation stability.

2. Literature Review

2.1 Multi-View Triangulation

The standard approach for recovering 3D structure from posed images is multi-view triangulation, where corresponding 2D points across two or more views are back-projected into 3D using known projection matrices. Given a projection matrix $\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$ for each view, the Direct Linear Transform (DLT) formulates an overdetermined homogeneous linear system $\mathbf{AX} = 0$ and solves via SVD [1]. OpenCV's `triangulatePoints` implements the two-view case efficiently. Since camera poses are provided in this project, we bypass Structure from Motion (SfM) pipelines such as COLMAP [2] and proceed directly to triangulation with the given extrinsics.

2.2 Oriented Bounding Box Fitting

Fitting a tight 3D bounding box to a point cloud is commonly done using Principal Component Analysis [3]. The three eigenvectors of the point cloud's covariance matrix define the box orientation, and extents are determined by projecting points onto each axis. A limitation of pure PCA is that the resulting axes may not align with the physical structure of the object. For a flat connector mounted on a vertical panel, one axis should be normal to the panel surface. We address this by explicitly estimating the panel surface normal from triangulated bounding box corners and assigning rotation axes accordingly.

2.3 Open-Vocabulary Object Detection

Vision-language models have made it possible to detect objects described in free-form natural language without task-specific training. CLIP [4] introduced contrastive image-text pre-training that enables zero-shot classification, though it lacks localization. OWL-ViT [5] extended this to open-vocabulary bounding box prediction. Grounding DINO [6] combines the DINO transformer backbone with grounded pre-training on paired image-text-box data, achieving state-of-the-art results on open-set detection benchmarks. Its ability to detect objects described by compound noun phrases such as *"IEC power inlet"* or *"RJ-45 ethernet port"* makes it directly applicable to our generalization task.

2.4 Neural Radiance Fields and 3D Gaussian Splatting

NeRF [7] and 3D Gaussian Splatting [8] represent the current state of the art for photorealistic novel view synthesis from posed images. Both learn a volumetric representation from multi-view supervision and render new views via differentiable ray-marching or rasterization respectively. Novel view synthesis was originally in scope for this project but was subsequently removed by the course instructor due to the compute resources required; our work focuses entirely on the semantic pose estimation component.

3. System Architecture and Design Rationale

The pipeline is organized into four sequential stages: data loading, semantic annotation, multi-view triangulation, and OBB fitting. A separate module handles the bonus generalization task by replacing the manual annotation stage with automatic Grounding DINO detection. The overall flow is:

Images + Poses + Intrinsics

$\downarrow$

Semantic Annotation (manual or Grounding DINO)

$\downarrow$

Multi-View Triangulation (dense grid, 400 pts/ROI)

$\downarrow$

OBB Fitting (PCA + panel normal + depth prior)

$\downarrow$

Output: `answers.json` *(center, extent, rotation)*

3.1 Baseline: Manual Annotation and Triangulation

For the known parts, we manually annotated 2D bounding boxes by visually inspecting the images at full 2560×1440 resolution. Frames 471 and 496 provide the clearest and most frontal views of the back panel and were selected as the annotation frames for all entities. Using only two frames per entity was deliberate: frames from very different viewpoints provide useful baseline variation for triangulation, while adding frames where the back panel is barely visible introduces noise without benefit.

Within each annotated bounding box, we sample a regular 20×20 grid (400 points). For each pair of annotated frames, corresponding grid points are matched by their normalized position within the bounding box—the point at relative position (t_x, t_y) in frame A corresponds to the same relative position in frame B. This is a reasonable assumption for a flat, fronto-parallel surface such as the PC back panel. Each matched pair is triangulated using `cv2.triangulatePoints`, and the resulting 3D point is retained if it has positive depth in both cameras and a reprojection error below 50 pixels.

We chose not to use SIFT feature matching within the ROI as an additional triangulation source because the back-panel surface is dark and nearly textureless, and SIFT tends to latch onto unreliable specularities that are inconsistent across views. The grid-based approach is more predictable for low-texture planar regions.

3.2 Bonus: Grounding DINO Detection

To handle new parts at evaluation time without re-annotation, we run Grounding DINO with a text prompt describing the requested component. The model processes all 16 frames and returns bounding box predictions with confidence scores. The top two detections from distinct frames are used as annotation inputs for the triangulation stage, keeping the generalization path architecturally identical to the baseline.

A detection threshold of 0.3 was used. Compound noun phrases with period-separated alternatives (e.g., *"USB port . USB connector . USB socket"*) consistently outperformed single-word prompts, likely because the model benefits from multiple surface forms of the same concept.

Table 1: Comparison of the two pipeline variants.

Aspect	Baseline (Manual)	Bonus (Grounding DINO)
Part detection	Manual 2D bounding boxes	Zero-shot text-prompt detection
Frames annotated	471, 496 (manually selected)	Top-2 by confidence, auto-selected
Triangulation	Dense grid, 400 pts/ROI	Dense grid, 400 pts/ROI
OBB fitting	PCA + panel normal + depth prior	PCA + panel normal + depth prior
New parts?	No — requires re-annotation	Yes — change text prompt only

4. Implementation Details

4.1 Camera Intrinsics and Coordinate Conventions

Camera intrinsics were taken from the provided `intrinsic.json`: $f_x = 1477.01$, $f_y = 1480.44$, $c_x = 1298.25$, $c_y = 686.82$, for image dimensions 2560×1440. The horizontal field of view is:

$$\text{HFoV} = 2\arctan\left(\frac{W}{2f_x}\right) = 2\arctan\left(\frac{1280}{1477}\right) \approx 82^\circ$$

An early attempt to estimate intrinsics using COLMAP feature extraction returned $f_x = f_y = 3072$, approximately twice the calibrated value. This would place reconstructed points at roughly half the correct depth and was therefore discarded. Camera poses are given as 4×4 camera-to-world matrices. Projection matrices are formed as:

$$\mathbf{P} = \mathbf{K} \cdot \mathbf{T}_{w2c}[:3,:] \quad \text{where} \quad \mathbf{T}_{w2c} = \mathbf{T}_{c2w}^{-1}$$

4.2 Triangulation and Point Cloud Cleaning

For each pair of annotated views, projection matrices $\mathbf{P}_1$ and $\mathbf{P}_2$ are constructed and passed to `cv2.triangulatePoints` with batched 2D correspondences. Each triangulated point is tested for positive depth in both cameras and for reprojection error below 50 pixels in either view.

After initial filtering, Median Absolute Deviation (MAD) outlier rejection is applied with a threshold of $3.0\,\hat{\sigma}$, where $\hat{\sigma} = 1.4826 \cdot \text{median}(|d_i - \tilde{d}|)$ is the MAD-based scale estimate. MAD is preferred over standard-deviation-based rejection because it is robust to the very outliers we wish to remove.

4.3 OBB Fitting with Panel Normal Estimation

OBB fitting proceeds in two steps. First, the panel surface normal is estimated by applying SVD directly to the mean-centered triangulated point cloud. For a near-planar set of points $\mathbf{C} \in \mathbb{R}^{N\times 3}$, the SVD $\mathbf{C} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$ gives the last row of $\mathbf{V}^T$ as the least-variance direction, which corresponds to the surface normal. This approach captures the actual geometric orientation of the back panel including its slight backward tilt in the Z direction.

Second, the three rotation rows are assigned following the GT evaluation convention: Row 2 is fixed as the world X axis (panel horizontal direction, confirmed from the VGA ground truth where Row 2 $\approx [1, 0, 0]$). Row 0 is the SVD-estimated surface normal with its X component zeroed to maintain perpendicularity with Row 2, and sign-corrected to ensure positive Y. Row 1 is then derived as $\mathbf{r}_1 = \mathbf{r}_2 \times \mathbf{r}_0$ and the system is re-orthogonalised for numerical stability.

Extents are computed from PCA spread of the inlier point cloud. The largest PCA axis gives the connector width, the medium axis gives the connector height, and the smallest axis (normal direction) is replaced by a physical depth prior of 6 mm—the approximate protrusion depth of a standard IEC or RJ-45 connector from a flat panel.

4.4 Manual Annotation Coordinates

Table 2: Manual 2D bounding box annotations (pixel coordinates, original resolution).

Entity	**Frame**	x_1	y_1	x_2	y_2
power_socket	471	1140	1020	1280	1100
power_socket	496	1670	1060	1820	1155
ethernet_socket	471	1160	305	1280	405
ethernet_socket	496	1735	305	1845	405
vga_socket	471	1101	357	1285	423
vga_socket	496	1671	364	1855	430

5. Experimental Results

5.1 Baseline OBB Estimates

Running the baseline pipeline on the annotated frames produces the OBB estimates shown in Table **??**. All coordinates are in the world frame defined by the provided poses.

Table 3: Estimated OBB parameters for all annotated entities (world frame, units in metres).

Entity	**Center** (x, y, z)	**Extent** (w, h, d)	**Triangulated pts**
power_socket	$(0.271,\ 0.207,\ 0.540)$	$(0.030,\ 0.018,\ 0.006)$	≈ 350
ethernet_socket	$(0.282,\ 0.225,\ 0.850)$	$(0.023,\ 0.018,\ 0.006)$	≈ 310
vga_socket	$(0.271,\ 0.226,\ 0.835)$	$(0.035,\ 0.012,\ 0.006)$	≈ 380

The estimated extents are physically reasonable. The power socket (IEC C14) is estimated at approximately 30×18 mm, close to the standard aperture of 29.2×12.8 mm. The ethernet socket width of approximately 23 mm is slightly wider than the RJ-45 standard aperture of 13.97 mm, likely because the bounding box includes the surrounding I/O shield opening rather than the connector body alone.

5.2 Validation Against VGA Ground Truth

The `sample_answers.json` file provides a complete OBB ground truth for the VGA socket. Table **??** summarises the comparison.

Table 4: Validation of VGA socket estimate against provided ground truth.

Metric	**Value**	**Assessment**
Center distance	≈ 0.6 mm	Sub-centimeter accuracy
Extent error (largest)	≈ 0.3 mm	Physically accurate
Rotation angular error	$\approx 6.8°$	Minor axis misalignment

The rotation error of approximately $6.8°$ is primarily a residual misalignment between the estimated panel normal direction and the true panel orientation. This likely originates from slight inaccuracies in the corner triangulation used to estimate panel directions—when the stereo baseline between frames 471 and 496 is not perfectly symmetric relative to the panel, the estimated normal picks up a small bias.

5.3 Grounding DINO Detection Performance

With a confidence threshold of 0.3, Grounding DINO produced the results shown in Table **??**.

Table 5: Grounding DINO detection results on the 16-frame dataset.

Text prompt	Detections / 16	Best conf.	Best frame
power socket . IEC connector . C13 inlet	14 / 16	0.57	frame_000390
ethernet port . RJ45 port . network port	16 / 16	0.50	frame_000353

Visual inspection confirmed that high-confidence detections (above 0.4) generally localize the correct region of the back panel. For the bonus pipeline, restricting to the top-2 detections from distinct frames effectively filters out the occasional mis-fires on other high-contrast image regions.

6. Conclusions

This project built a complete pipeline for metric-semantic pose estimation of PC back-panel components from a sparse set of 16 posed images. The approach combines classical multi-view triangulation with a physically-informed OBB fitting strategy, and extends to open-vocabulary detection for generalization to unseen parts.

Several aspects proved important in practice. First, using calibrated intrinsics rather than COLMAP-estimated ones was essential—the COLMAP estimate was off by a factor of approximately $2\times$, which would have produced large depth errors. Second, the grid-based correspondence approach, while simple, worked well for the flat, low-texture back-panel surface where SIFT feature matching is unreliable. Third, explicitly estimating the panel surface normal for OBB rotation assignment gave results better aligned with the evaluation convention than pure PCA alone.

The main limitation is the reliance on only two frames per entity for triangulation. With more annotated frames or a better stereo baseline, triangulation quality and coverage would improve. The $6.8°$ rotation error in the VGA validation suggests room to improve the panel normal estimation, perhaps by incorporating more views or fitting a plane directly to the full reconstructed point cloud.

7. Bonus: New Ideas Explored

Beyond the core pipeline, two additional directions were explored.

(a) 3D Gaussian Splatting for novel view synthesis. We attempted to train a 3DGS model using nerfstudio's *splatfacto* implementation on the provided dataset. The main challenges were: (i) the gsplat CUDA extension required over 20 minutes to compile on a T4 GPU and repeatedly caused memory exhaustion during compilation; (ii) splatfacto requires a sparse point cloud for Gaussian initialization, which we generated by running COLMAP on the 16 frames and converting the output to PLY format; (iii) random initialization as an alternative resulted in slower convergence and lower quality at 30k iterations. We deprioritized this component when the novel view synthesis requirement was removed from the project specification.

(b) Automated intrinsic estimation via COLMAP. We investigated using COLMAP's feature extractor in CPU-only mode (to avoid the OpenGL dependency on headless servers) to recover camera intrinsics from the images themselves. COLMAP returned $f_x = f_y = 3072$ with zero distortion, suggesting it was unable to fully disambiguate focal length from scene scale without known 3D reference points. This experiment motivated our switch to the calibrated intrinsics from `intrinsic.json` and highlighted a practical pitfall of relying on auto-calibration for short-baseline multi-view datasets.

References

[1] Hartley, R., & Zisserman, A. (2004). *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.

[2] Schönberger, J. L., & Frahm, J. M. (2016). Structure-from-motion revisited. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[3] Gottschalk, S., Lin, M. C., & Manocha, D. (1996). OBBTree: A hierarchical structure for rapid interference detection. In *Proceedings of SIGGRAPH*.

[4] Radford, A., et al. (2021). Learning transferable visual models from natural language supervision. In *Proceedings of ICML*.

[5] Minderer, M., et al. (2022). Simple open-vocabulary object detection. In *Proceedings of ECCV*.

[6] Liu, S., et al. (2023). Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. *arXiv preprint arXiv:2303.05499*.

[7] Mildenhall, B., et al. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. In *Proceedings of ECCV*.

[8] Kerbl, B., et al. (2023). 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics* (SIGGRAPH).